6,906: Economics: Macro for Computer Science

Lecture 5: Facts about business cycles

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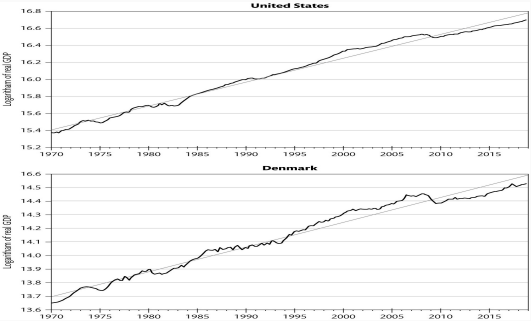
Learning objectives

- By the end of this lecture, you should be able to:
  - Characterize the fluctuations in economic activity about the long-term trend, known as "business cycles."

Definition of business cycles



Source: Sorensen and Whitta-Jacobsen (2022)

- In Part II, we focused on the long-run growth of output and proposed reasons why advanced economies have seen their real GDP trending up over time.
- However, when looking at the plot of the natural logarithm of real GDP we observe that the growth rate of the economy is not constant over time (if that were the case, the log of real GDP would follow the straight line).

Definition of business cycles

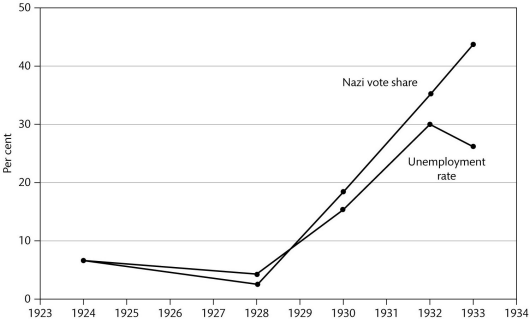
- In the short and medium run, the growth rate of real GDP fluctuates considerably, there being periods with higher or lower (even negative) growth rates than the average.
- These fluctuations in real GDP about the long-term trend are called "business cycles."
- The periods characterized by rapid growth are called business cycle "expansions" or "booms", while the periods characterized by declining economic activity are called business cycle "contractions" or "recessions."
- Note that the GDP series are seasonally adjusted, therefore the fluctuations are not due to seasonal swings in activity but to more fundamental causes that we will further study in the course.

Characteristics of the business cycle

- While the movements in real GDP are usually used as an indicator of the state of the economy, business cycles are characterized by a co-movement of a large number of economic variables.
- A full business cycle lasts more than a year, which indicates that business cycle movements display persistence.
- Business cycles occur in decentralized market economies.
- Although business cycles are recurrent, they are not periodic. The duration of a business cycle varies from slightly more than a year to 8-10 years, while the severity also varies considerably, with some recessions turning into depressions.

Why understanding business cycles is important

- On several occasions in history, recessions have developed into severe economic depressions, paving the way for social and political disaster.
- The figure shows a strong correlation between the unemployment rate in interwar Germany and the votes in favor of the Adolf Hitler's Nazi Party.
- Even though there were several other factors explaining Hitler's rise to power, there is no doubt that the Great Depression of the 1930s made it easier for him to gather support.



Source: Sorensen and Whitta-Jacobsen (2022)

Why understanding business cycles is important

- While depressions are exceptional events, there are many recent examples of economic downturns and, without exception, they caused economic welfare losses.
- Recessions usually bring considerable economic hardship to workers who lose their jobs, businesses that go bankrupt, and households that suffer capital losses on their assets. Even those not directly affected by bankruptcies and layoffs suffer from economic uncertainty and fears of future income and wealth losses.
- For these reasons, understanding business cycles is important because it may determine stabilization policy actions that reduce business cycles and improve the confidence of economic agents.

Measuring business cycles

- As our focus is on movements in real GDP and other aggregate variables about the long-run trend, we first need to remove a trend from the observed data.
- Let's assume that a series can be decomposed into a "trend'' component, denoted with a superscript "g'', and a cyclical component, denoted with a superscript "c''.
- Suppose the series is log real GDP. The decomposition of real GDP into its trend and cyclical component is given by:

$$lnY_t = lnY_t^g + lnY_t^c$$

- Our objective is to come up with a time series of the trend component, since, once we have that, the cyclical component is simply defined as the residual, i.e. $lnY_t^c = lnY_t - lnY_t^g$.

Trends and cycles

- There are several approaches to remove a trend from a trending time series.
  - The most obvious one is to fit a straight line through the series, the straight line being the "trend" component while the deviations of the actual series from the trend would be the cyclical component. However, this implies that the economy would always be in an equilibrium with constant growth if it were not disturbed by business fluctuations, while the theory of economic growth does not give us any reason to believe that the economy is always in a steady state.
  - Another way to isolate the trend component would be to take a moving average, where the trend component would be at any point in time the average realization of the actual time series in a "window" around that time. For example, for quarterly time series, a two-sided moving average measure of the trend would be

$$lnY_t^g = average(lnY_{t-4}, lnY_{t-3}, lnY_{t-2}, lnY_{t-1}, lnY_t, lnY_{t+1}, lnY_{t+2}, lnY_{t+3}, lnY_{t+4})$$

HP filter

- One method that allows for variation over time in the growth trend, while ensuring that short-term fluctuations are categorized as temporary cyclical deviations from the trend, which became widely used is the Hodrick Prescott (HP) filter.
- The basic idea would be to apply some sort of moving average to obtain the trend component of the series. Then subtract this from the original series to recover the cyclical component.

HP filter

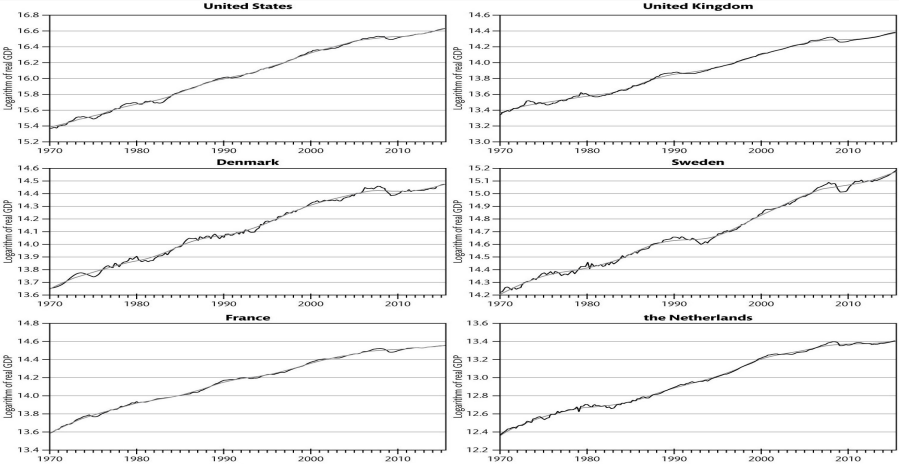
- The HP filter removes the trend component $y_t^g \equiv lnY_t^g$ from a time series $y_t \equiv lnY_t$ by solving the following minimization problem:

$$\min \sum_{t=1}^{T} \left[(y_t - y_t^g)^2 + \lambda((y_{t+1}^g - y_t^g) - (y_t^g - y_{t-1}^g))^2 \right] \tag{1}$$

  with respect to y_t^g.
- The first term aims to get y_t^g as close as possible to the observed series y_t, while the second term minimizes the changes in the trend series over time.
- The smoothing parameter λ penalizes fluctuations in the growth of the trend component:
  - if $\lambda = 0$, $y_t = y_t^g$ (no smoothing);
  - if $\lambda \to \infty$, y_t^g is a straight line;
  - with quarterly data λ is often set equal to 1600.

HP filter

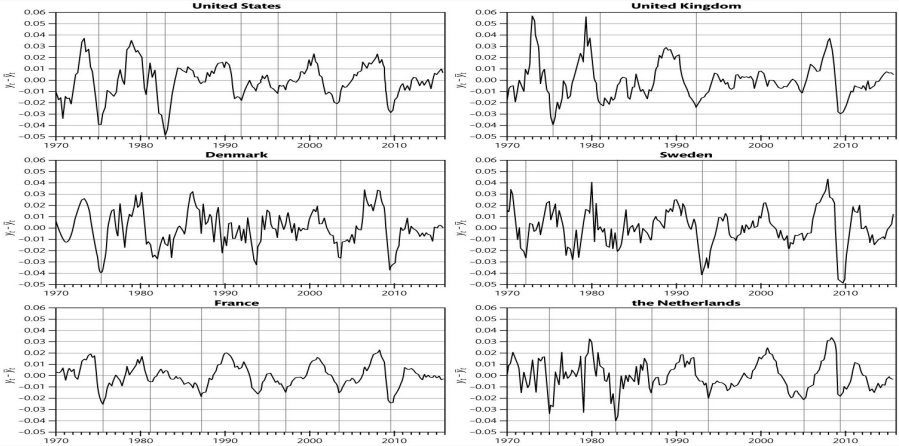
Real GDP and its trend obtained with the HP filter ($\lambda = 1600$):



Source: Sorensen and Whitta-Jacobsen (2022)

HP filter

Cycles in real GDP obtained with the HP filter ($\lambda = 1600$):



Source: Sorensen and Whitta-Jacobsen (2022)

Note that the vertical lines indicate business cycles measured from trough to trough.

Troughs and peaks

- Identification of business cycles troughs and peaks is based on the following simple rule of thumb:
  - A trough must be followed by a peak, and a peak must be followed by a trough.
  - The expansion phase as well as the contraction phase must last for a minimum of two quarters.
  - A business cycle must span a minimum of five quarters.

Stylized US business cycle facts

- We will look at some stylized facts about the US business cycles.
- Specifically, what we will do is compute "business cycle moments."
- "Business cycle moments" focus primarily on a variety of second moments such as:
  - volatility of a stationary series - its standard deviation;
  - cyclicality of a stationary series - its contemporaneous correlation with output;
    - if a series is positively correlated with output, we say that series is procyclical.
    - if a series is negatively correlated with output, we say that series is countercyclical.
    - if a series is roughly uncorrelated with output, we say that it is acyclical.
  - persistence of a stationary series - its first-order autocorrelation;

Stylized US business cycle facts

- For this discussion, we use US data for the period 1955Q1:2019Q4.
- All series are real, in logs, and expressed in per capita terms.

Series	**Std. Dev.**	**Rel. Std. Dev.**	**Autocorr.**	**Corr. w/ Y_t**
Output	0.0143	1.0000	0.8475	1.0000
Consumption	0.0080	0.5623	0.8316	0.7899
Investment	0.0538	3.7603	0.8495	0.9218
Hours worked	0.0178	1.2440	0.9120	0.8552
Wages	0.0094	0.6555	0.7009	0.1932

Stylized US business cycle facts

- Consumption is significantly less volatile than output; in contrast, investment is more than 3.5 times more volatile than output. Investment is the most unstable component of aggregate demand.
- Hours worked are about as volatile as output, while typically employment is less volatile over the business cycle than GDP.
- Wages are less volatile than output.
- Consumption, investment, and hours worked are strongly procyclical.
- Wages are weakly correlated with output.
- Almost all HP-filtered series are strongly persistent.
- There is considerable persistence in GDP and about the same degree of persistence in consumption.
- Hours worked tend to be even more persistent than GDP.

Summary

- In this lecture we learned:
  - To describe some facts about business cycles.
- Next lecture we begin the study of the macroeconomic model that may explain the short- and medium-run fluctuations in economic variables by first analyzing the goods and financial markets.